

Addition of angular momentum 1

For orbital angular momentum $\mathbf{L}$ we have

$$L_x\psi = -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \psi$$

$$L_y\psi = -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \psi$$

$$L_z\psi = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \psi$$

and for spin angular momentum $\mathbf{S}$

$$S_x\chi = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \chi$$

$$S_y\chi = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \chi$$

$$S_z\chi = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \chi$$

Let $\mathbf{J}$ be total angular momentum

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

Let Ψ be the product of wave function ψ and spin vector χ .

$$\Psi = \psi\chi$$

Then

$$\mathbf{J}\Psi = \mathbf{L}\Psi + \mathbf{S}\Psi$$

Let J^2 be magnitude-squared of total angular momentum.

$$J^2 = \mathbf{J} \cdot \mathbf{J} = J_x^2 + J_y^2 + J_z^2$$

Operator J^2 can be decomposed as

$$J^2 = (\mathbf{L} + \mathbf{S}) \cdot (\mathbf{L} + \mathbf{S}) = L^2 + S^2 + 2\mathbf{L} \cdot \mathbf{S} \quad (1)$$

Verify J^2 commutes with L^2 and S^2 .

$$[J^2, L^2] = [J^2, S^2] = 0$$

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